



VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956



Module-1

Unit-2

Experiment Tracking, Data and Model Management



Contents

- **Feature Stores**
- **Concept**
- **Benefits**
- **Role in AI applications**
- **Feast overview**



Feature Stores

- A feature is simply an input given to a model.
- A **feature store** is a system used to store, manage, share, and reuse the input features used by machine-learning models.
- Instead of calculating these features separately every time a model is trained, they can be created once and stored for reuse.





Feature Stores

- **Why Do We Need a Feature Store?**
- Consider two teams developing different models.
 - **Team A**
 - Placement prediction model.
 - Uses:
 - CGPA
 - coding score
 - attendance percentage
 - internship count
 - **Team B**
 - Student fail-risk prediction model.
 - Also needs:
 - CGPA
 - attendance percentage
 - assignment average

“Without a feature store, both teams independently calculate the same features”



Feature Stores

- **Why Do We Need a Feature Store?**
- This can create:
 - duplicated work
 - different formulas
 - inconsistent values
 - difficult maintenance

*Raw Data -> Feature Engineering -> Feature Store -> Placement Model,
Student Fail-Risk Model, Recommendation Model*





Feature Stores

- **Feature Engineering vs Feature Store**
- **Feature Engineering**
 - Creating useful information from raw data.
- **Feature Store**
- Storing and managing the created feature.



Feature Stores

- **Offline and Online Feature Stores**
- Feature stores often have two uses.
- **Offline Features**
- Used mainly for:
 - historical data
 - model training
 - batch analysis

“Student attendance for the last four semesters”



Feature Stores

- **Offline and Online Feature Stores**
- Feature stores often have two uses.
- **Online Features**
- Used for immediate predictions.
- Example:
- A fraud-detection system may need
“Number of transactions in the last 10 minutes”

Offline Store → Training

Online Store → Real-time Prediction



Feature Stores

- **What Does a Feature Store Maintain?**
- A feature store may maintain:
 - feature name
 - feature value
 - data type
 - entity ID
 - feature definition
 - source
 - timestamp
 - version
 - owner
 - description

Feature	Entity	Value	Updated
attendance_percentage	S101	84	2026-08-01
coding_score	S101	76	2026-08-01
internship_count	S101	2	2026-08-01



Feast

- **Feast is an open-source feature store for machine-learning systems.**
- Its purpose is to make features available consistently for:
 - model training
 - batch prediction
 - real-time prediction
- Feast manages feature definitions and retrieval.
- It does **not** train or deploy the ML model itself.

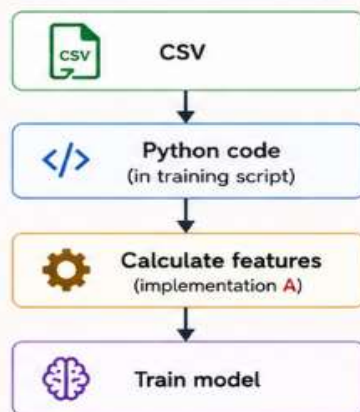


Feast

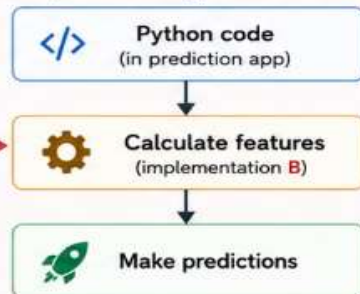
WITHOUT FEAST

Features are calculated inside each application.

Risk: Inconsistent features across training and prediction.



Later, another application...

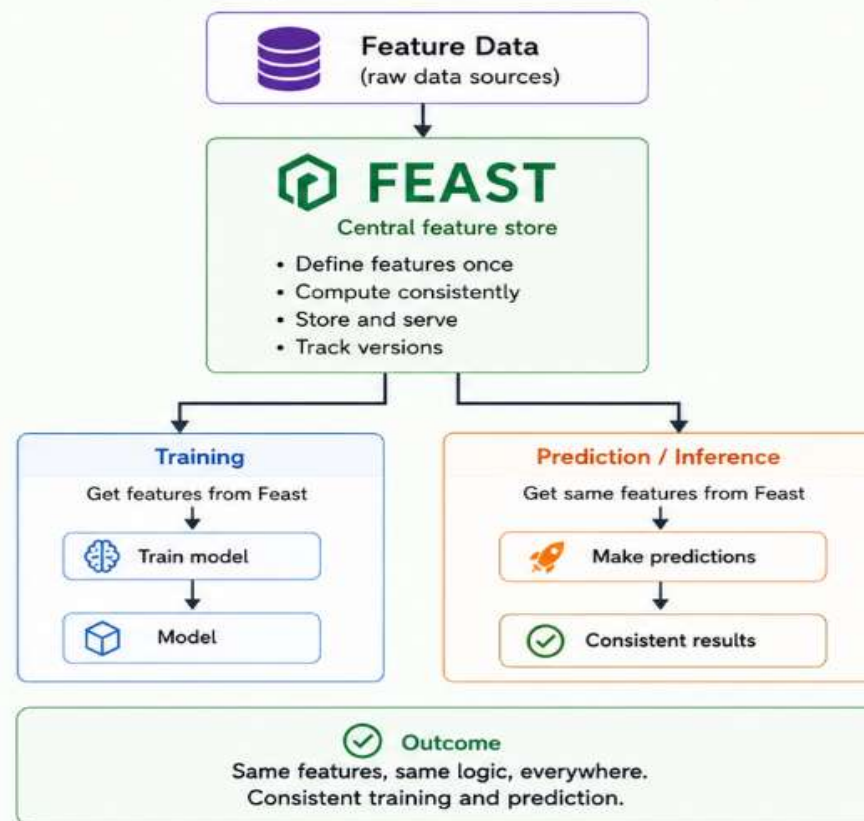


Problem

Same features are calculated in a different way. This can lead to different results.

WITH FEAST

Features are defined once, stored, and reused everywhere.
Benefit: Consistent, reliable features for training and prediction.

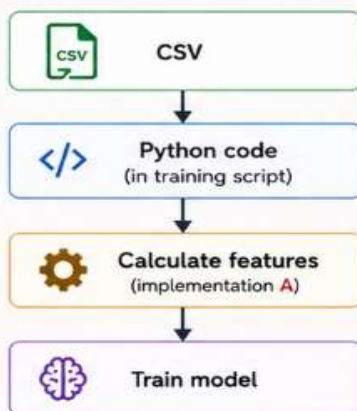


Feast

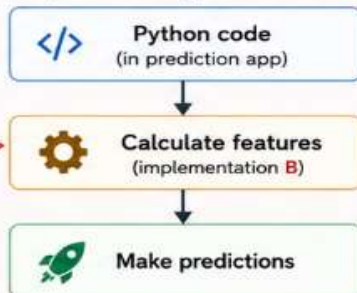
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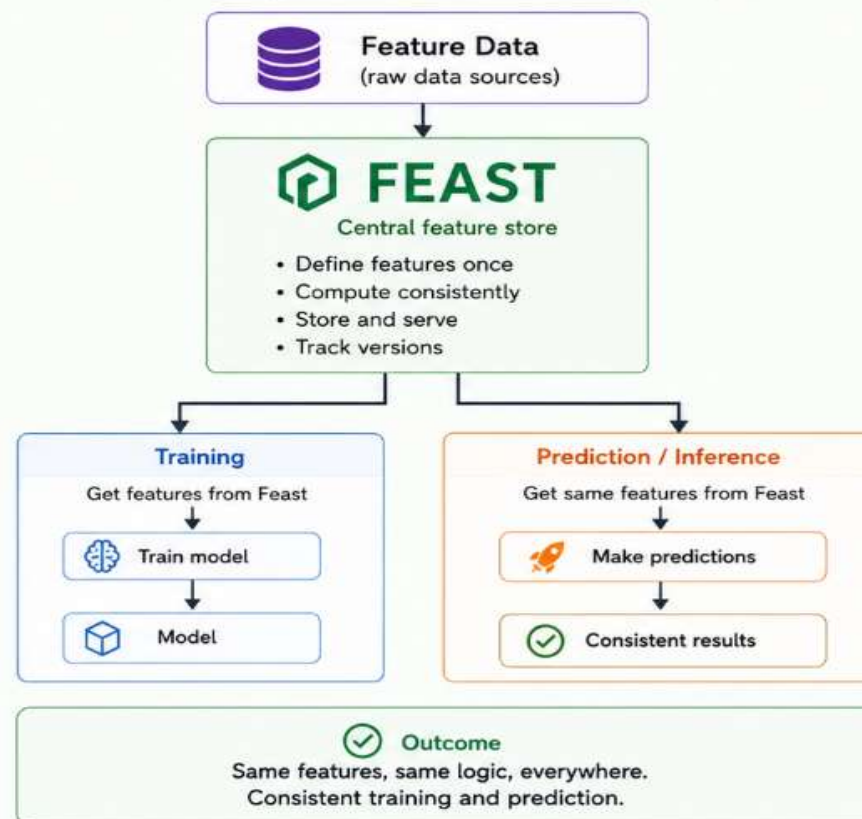


Problem

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WITH FEAST

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Feast

- **Feast Concepts**

- **Entity**

- An **entity** identifies the object for which features are stored.

Application	Entity
Student placement	Student
Fraud detection	Customer
House-price prediction	House
Product recommendation	Customer or Product
Skill-gap analysis	Skill



Feast

- **Feast Concepts**
- **Feature**
- A feature is a value associated with an entity.
 - For the entity “Skill”
 - Skill = Docker
 - Possible features:
 - industry_demand = 0.72
 - job_role_count = 5
 - course_count = 1
 - practical_coverage = 0
 - project_coverage = 0



Feast

- **Feast Concepts**
- **Feature View**
- A **FeatureView** logically groups related features and connects them to a data source. Feast uses feature views when building historical training datasets and when loading/retrieving online features.

Industry Feature View

industry_demand
job_frequency
role_count

Curriculum Feature View

course_count
practical_coverage
assessment_coverage
project_coverage





Feast

- **Feast Concepts**
- **Registry**
- The Feast Registry stores definitions of objects such as:
 - data sources
 - feature views
 - feature services
- It allows Feast to know what features exist and how they are organized.





Feast

- **Feast Concepts**
- **Feature Service**
- A **Feature Service** defines a collection of features required by a particular model or model version.
- Suppose our skill gap-classification model needs
 - industry_demand
 - role_count
 - course_count
 - practical_coverage
 - project_coverage
- These can be grouped as:
 - Skill Gap Model Feature Service





Feast

- **Feast Operation**
- Suppose features currently exist in historical storage.
- Offline Store -> Materialize -> Online Store
- **Materialization** copies feature values needed for online serving from the offline data source into the online store.
- This allows the model to access current feature values quickly.



Feast

- Feast Operation





Thank You

